JOHNSON CONTROLS, INC.
14 Woodcross Drive
Columbia, SC 29212
(803)781-7642

Report of Eddy Current Inspection

Manufacturer: York

Model: YTK3K3

Serial: YMXM604029

Location: OAK MITSUI
29 BATTLESHIP ROAD
CAMDEN, SC 29020

Inspected: December 17, 2013

Inspected By: LARRY B. WARNOCK, LEVEL III

Reviewed By: ______________________
TECHNICAL MANAGER, LEVEL III

Table of Contents

Section A . . . . Inspection Report

Part I . . . . Vessel Information

Part II . . . . Summary of Inspection

Part III . . . . Defect Summary

Part IV . . . . Recommendations

Part V . . . . Data Sheets

Part VI . . . . Tube Chart Layout

Part VII . . . . Tube Strip Charts

Section B . . . . Calibration Settings and Graphs

Calibration Procedures

Section C . . . . Explanation of Abbreviations

Vessel Information

Manufacturer	Model	Style	Serial Number	Type
York	YTK3K3	Open Drive	YMXM604029	Centrifugal

Condenser	
TestEnd	Opposite Inlet/Outlet
Tube Count	550
Tube Type	Skip Fin IE
Tube Material	Copper
OD	.750
*NWT/Under Fins	.035
*NWT/Bell/Land	.052
#/Type Support	3 Mild Steel
Tube Numbering	Top to Bottom
Row Numbering	Left to Right
Tube Length +- 2	156 Inches

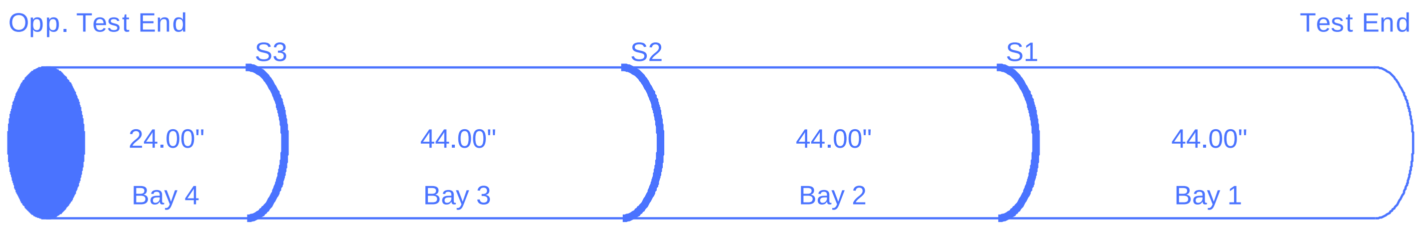
Evaporator	
TestEnd	Outlet End
Tube Count	618
Tube Type	Skip Fin IE
Tube Material	Copper
OD	.750
*NWT/Under Fins	.028
*NWT/Bell/Land	.049
#/Type Support	3 Mild Steel
Tube Numbering	Left to Right
Row Numbering	Top to Bottom
Tube Length +- 2	156 Inches

Analyst: LARRY B. WARNOCK, LEVEL III

* Nominal Wall Thickness

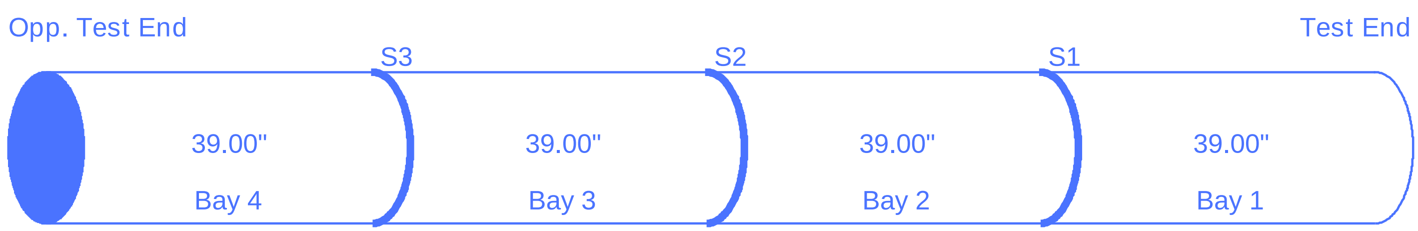
Vessel Bay Length Information

Condenser (Length = 156 inches)
S = Intermediate Support



Bay 4	24.00"
Bay 3	44.00"
Bay 2	44.00"
Bay 1	44.00"

Evaporator (Length = 156 inches)
S = Intermediate Support



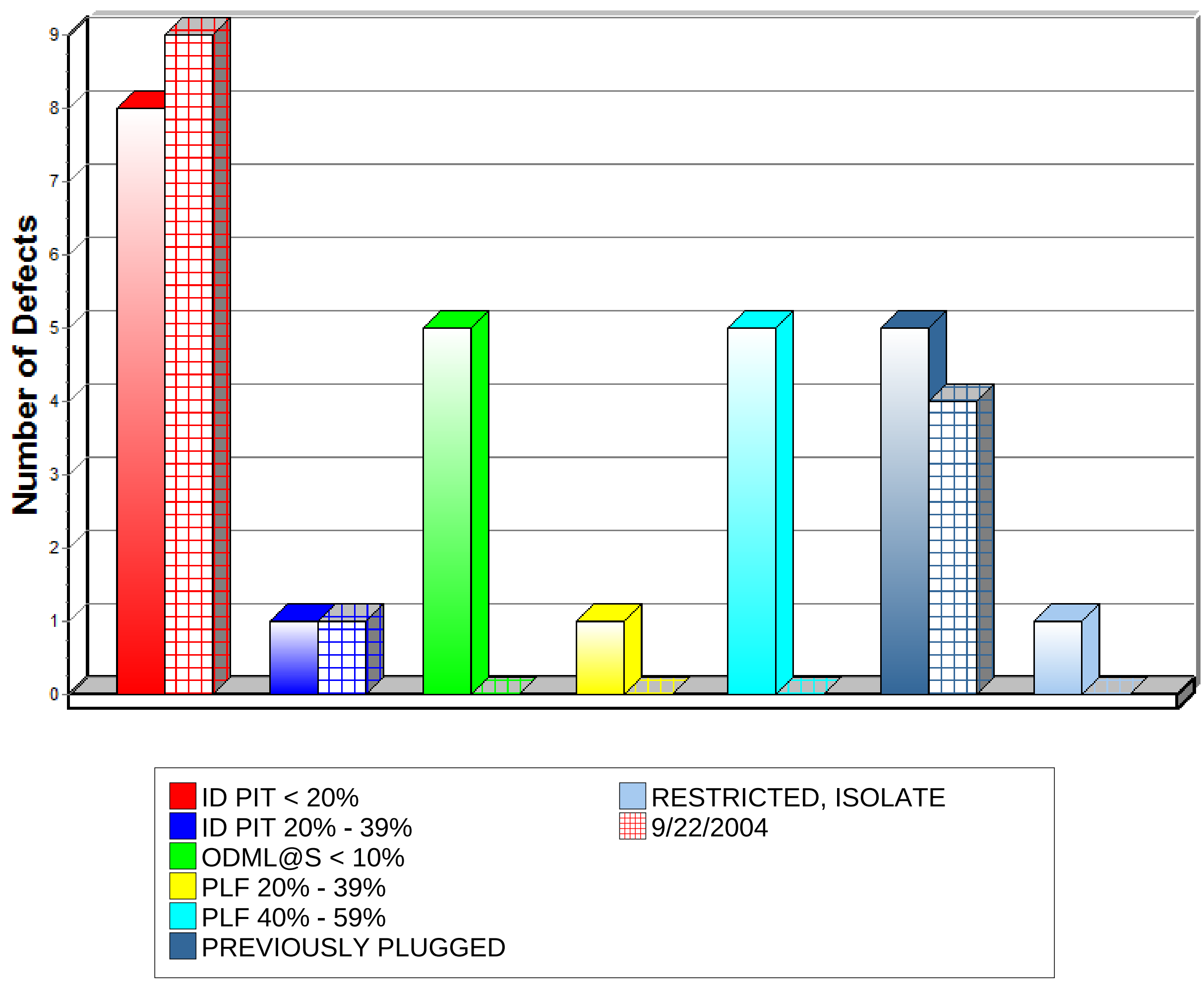
Bay 4	39.00"
Bay 3	39.00"
Bay 2	39.00"
Bay 1	39.00"

Defect Summary/Comparison

Comparison of Tests Performed

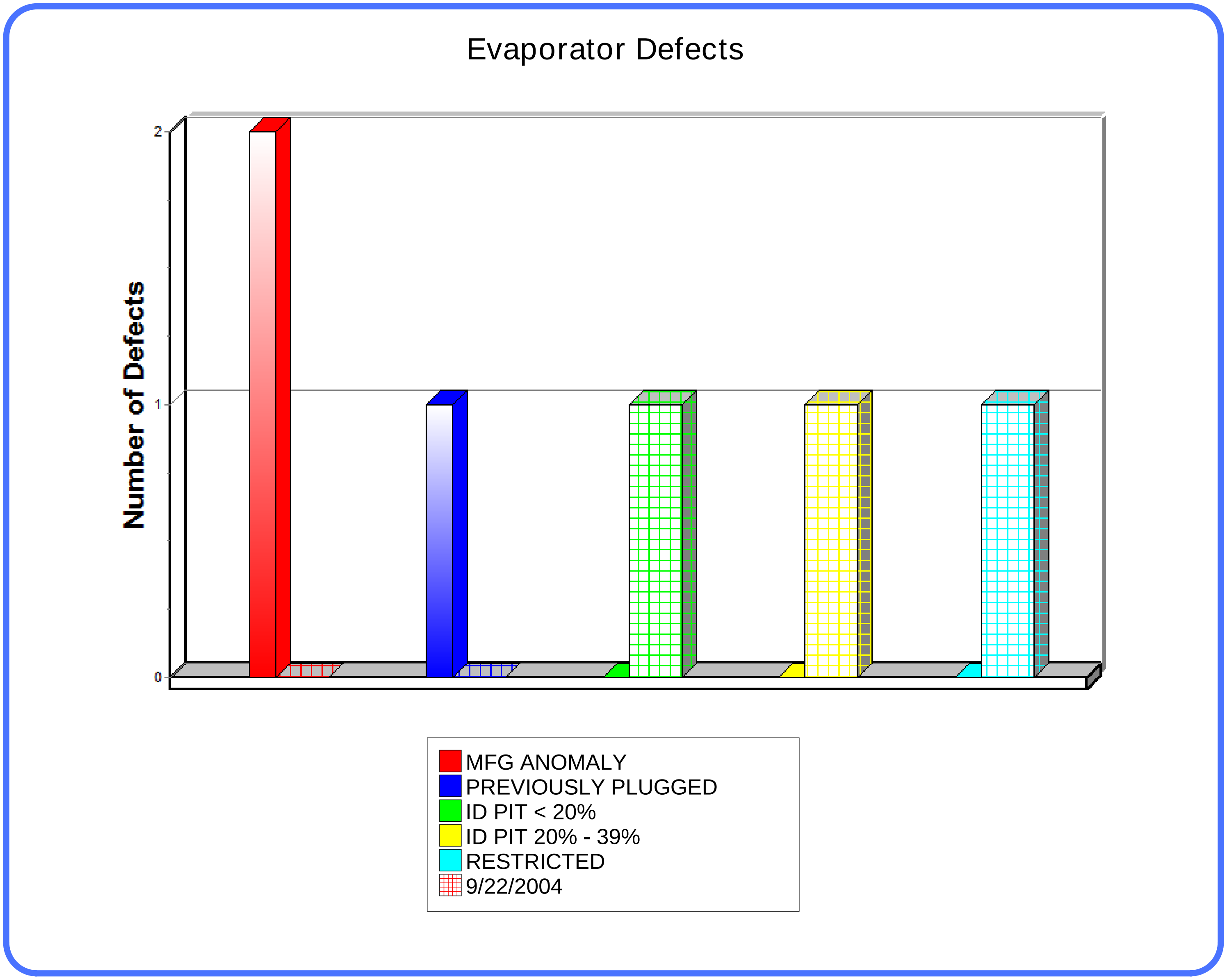
12/17/2013 9/22/2004

Condenser Defects



Location	Model	Serial Number
OAK MITSUI	YTK3K3	YMXM604029

Note: The Graph will indicate a Comparison Analysis when the unit has been previously tested by TAI Services.



Location	Model	Serial Number
OAK MITSUI	YTK3K3	YMXM604029

Note: The Graph will indicate a Comparison Analysis when the unit has been previously tested by TAI Services.

Summary of Inspection

An eddy current tube inspection was performed as part of a preventive maintenance program with the following results.

Condenser: 550 Tubes		
Significant/Measurable Indications	Number of Tubes Marked	Percent of Bundle
ODML@S < 10%	5	.91
PLF 20% - 39%	1	.18
PLF 40% - 59%*	5	.91
ID PIT < 20%	8	1.45
ID PIT 20% - 39%	1	.18
PREVIOUSLY PLUGGED	5	.91
RESTRICTED, ISOLATE*	1	.18
Totals	26	4.72

* REQUIRES ACTION

Evaporator: 618 Tubes		
Significant/Measurable Indications	Number of Tubes Marked	Percent of Bundle
MFG ANOMALY	2	.32
PREVIOUSLY PLUGGED	1	.16
Totals	3	.48

Recommendations

An eddy current inspection was performed on the tubes in this machine. This test was performed using accepted eddy current test methods for the inspection of in-service tubing. It should be noted that Eddy Current is not a leak detection method. The possibility does exist that tubes could contain defects and/or leaks which are not detectable. If leaks are suspected, we recommend a pressure test be used to identify the leaking tubes.

The following suggested repair actions are based on accepted industry standards. After removing sample tubes to confirm the inspection results, a determination of corrective action should be made by the repair agency and end user. Only these parties have knowledge of the critical applications and long-term use of the equipment. If plugging is selected over replacement, both efficiency and capacity should be considered.

CONDENSER:

Tubes indicating OD Metal Loss at a Support require no corrective action at this time. This type damage is typically caused by corrosion resulting from the formation of acids in the Refrigerant. Loss at a support can also be caused by tube vibration.

We recommend tubes indicating Possible Longitudinal Flaws (PLF) of 40% or more should be isolated from the system. Tubes indicating possible longitudinal flaws of less than 40% usually require no corrective action but should be monitored for defect growth.

Tubes indicating ID Pits require no corrective action at this time. However, this type damage should be monitored for defect growth, as it can be progressive.

Tubes marked as Previously Plugged, had been plugged prior to this inspection.

We recommend tubes marked as Restricted, Isolate, be isolated from the system at this time. It is believed these tubes contain defects or damage which prevented probe access.

EVAPORATOR:

The tubes indicating Abnormal Nominal Indications require no corrective action at this time. The tubes indicate possible Manufacturing Anomalies. The exact cause of the indication is unknown. However, it is believed to be a non-progressive manufacturing anomaly, and should not affect the performance or longevity of the affected tubes.

Tubes marked as Previously Plugged, had been plugged prior to this inspection.

RE-INSPECTION RECOMMENDATIONS:

We recommend that a follow-up inspection be performed on these vessels as follows:

Condenser: 17 December 2016

Evaporator: 17 December 2016

A copy of this report should be retained in your files to be used for comparison at that time.

If you should have any questions concerning this report, or if we may be of further assistance, please feel free to call upon us.

Data Sheet

Location	Model	Serial Number	Date
OAK MITSUI	YTK3K3	YMXM604029	December 17, 2013
29 BATTLESHIP ROAD			
CAMDEN, SC 29020			

Row	Tube	Description	Area	Action Req.
		SET UP CALIBRATE & STARTED		
		CONDENSER 12/17/2013 08:04 am		
4	13	ID PIT < 20%	B01	
6	2	ID PIT < 20%	B01	
11	12	ID PIT < 20%	B03	
14	9	ID PIT < 20%	B02	
16	1	ODML@S < 10%	S01	
16	2	PREVIOUSLY PLUGGED	TE	
16	3	PREVIOUSLY PLUGGED	TE	
17	1	PREVIOUSLY PLUGGED	TE	
17	2	ODML@S < 10%	S01	
17	6	PREVIOUSLY PLUGGED	TE	
18	1	PREVIOUSLY PLUGGED	TE	
18	2	ODML@S < 10%	S01	
19	1	ODML@S < 10%	S01	
19	3	PLF 40% - 59%	B04	✓
20	1	ODML@S < 10%	S01	
20	3	ID PIT < 20%	B04	
21	3	PLF 40% - 59%	B02	✓
21	4	PLF 40% - 59%	B02	✓
21	9	PLF 40% - 59%	B02	✓

Row	Tube	Description	Area	Action Req.
23	6	RESTRICTED, ISOLATE	B04	✓
24	16	ID PIT < 20%	B04	
25	16	ID PIT 20% - 39%	B02	
27	16	PLF 40% - 59%	B02	✓
28	15	ID PIT < 20%	B04	
30	11	PLF 20% - 39%	B03	
30	12	ID PIT < 20%	B04	
CALIBRATION CHECK & COMPLETED				
CONDENSER 12/17/2013 11:18 am				
SET UP CALIBRATE & STARTED				
EVAPORATOR 12/17/2013 11:35 am				
11	14	MFG ANOMALY	LL	
17	25	PREVIOUSLY PLUGGED	TE	
18	15	MFG ANOMALY	02	
CALIBRATION CHECK & COMPLETED				
EVAPORATOR 12/17/2013 03:26 pm				

Condenser Section

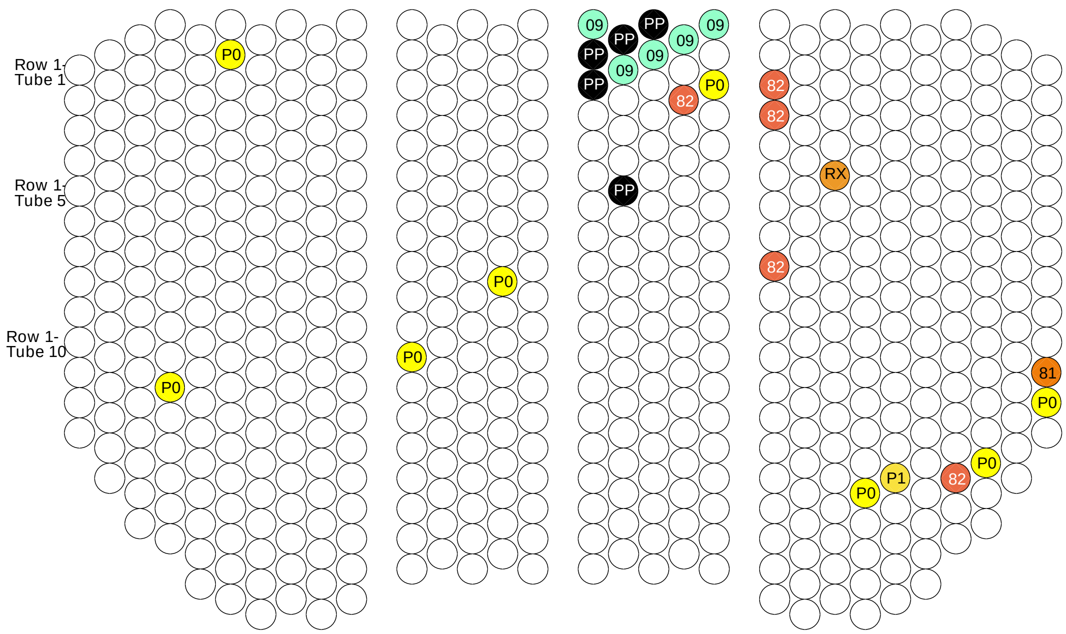
S/N YMXM604029

Opposite Inlet/Outlet

Rows numbered
Left to Right

Tubes numbered
Top to Bottom

Top of Vessel



09 = ODML@S < 10%

81 = PLF 20% - 39%

82 = PLF 40% - 59% REQUIRES ACTION

P0 = ID PIT < 20%

P1 = ID PIT 20% - 39%

PP = PREVIOUSLY PLUGGED

RX = RESTRICTED, ISOLATE REQUIRES ACTION

Evaporator Section

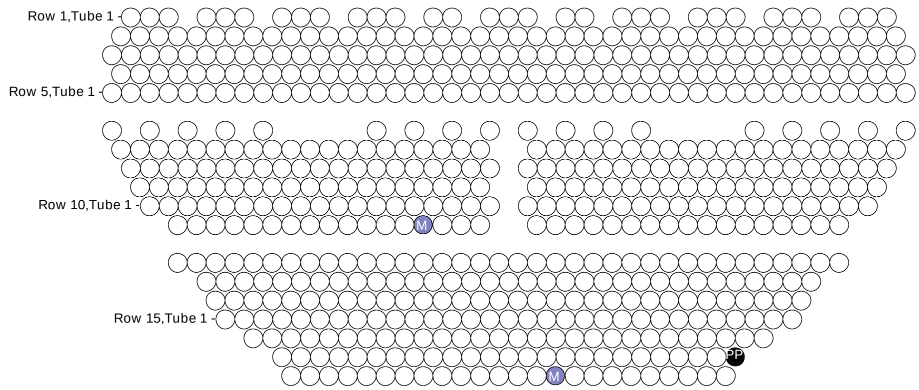
S/N YMXM604029

Outlet End

Tubes Numbered
Left to Right

Rows Numbered
Top to Bottom

Top of Vessel



M = MFG ANOMALY

PP = PREVIOUSLY PLUGGED

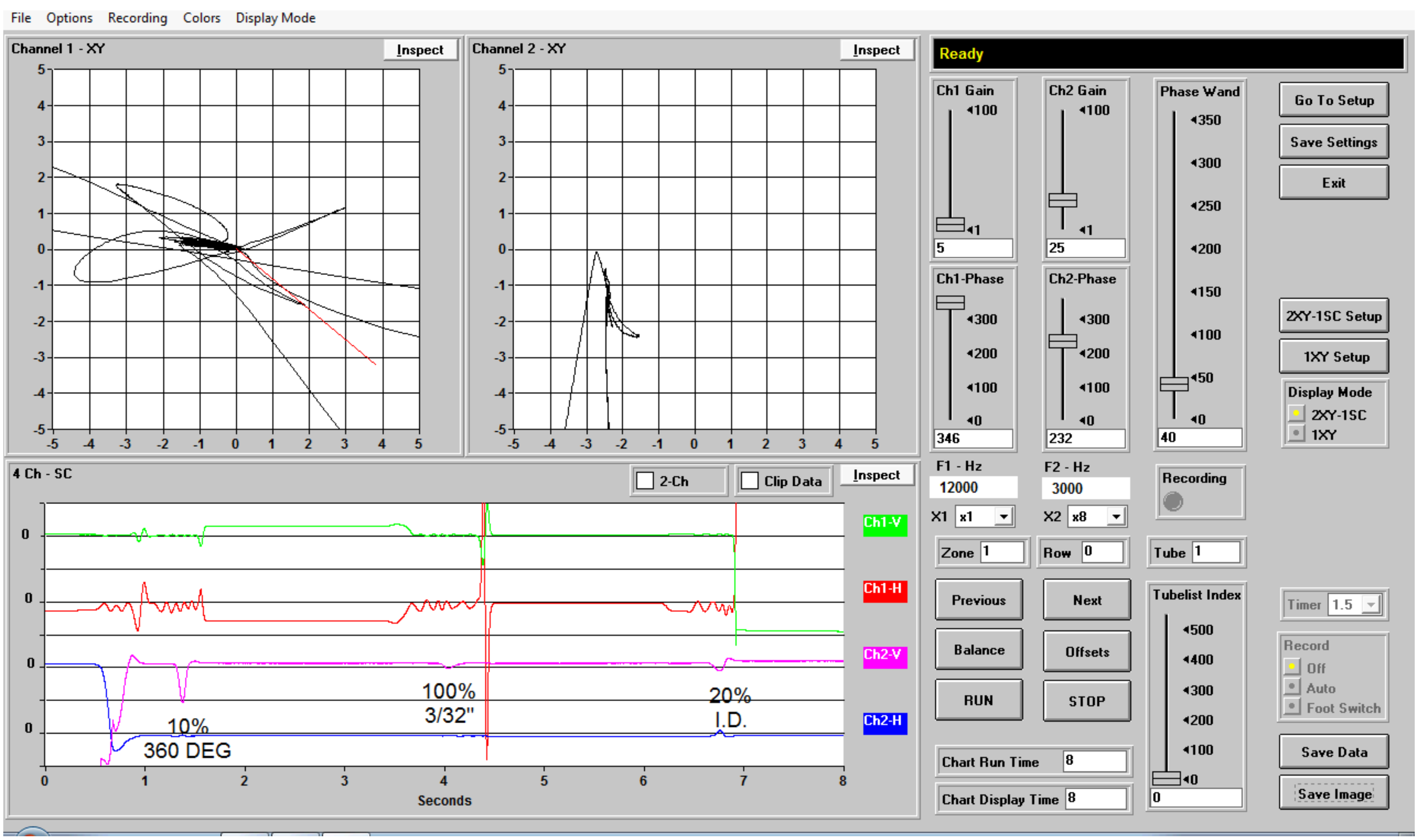
Test Instrument Used
Highline Automation, EC14

S/N YMXM604029

Calibration Page

Tube Type	Material	Nom Wall Thick	End Wall Thick	OD	Test Type	Probe Diameter
Skip Fin IE	Copper	.035	.052	.750	CROSS/DIFF	.500

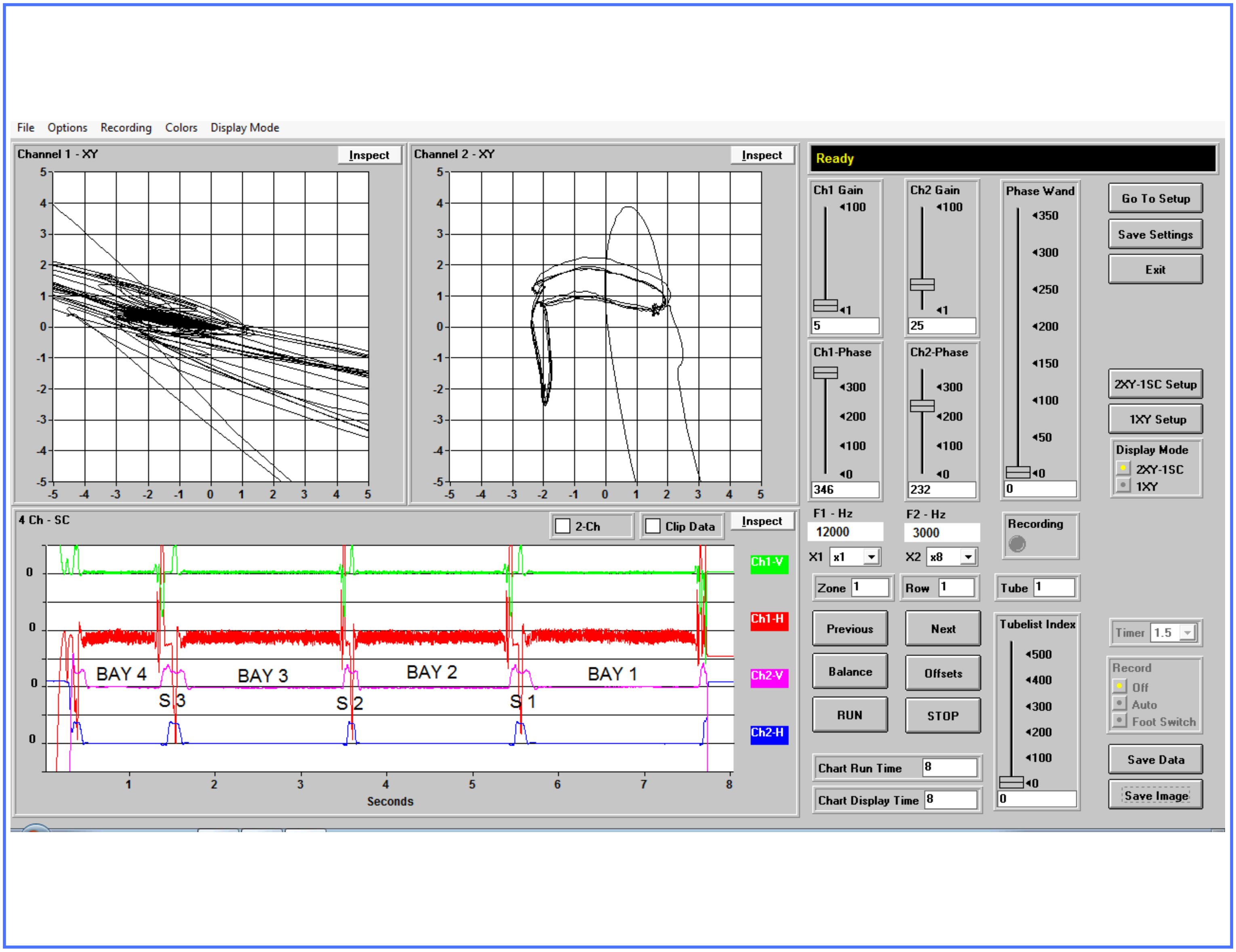
Condenser



Note: Defects are compared to machined standards.
Actual Defect Geometry may differ.

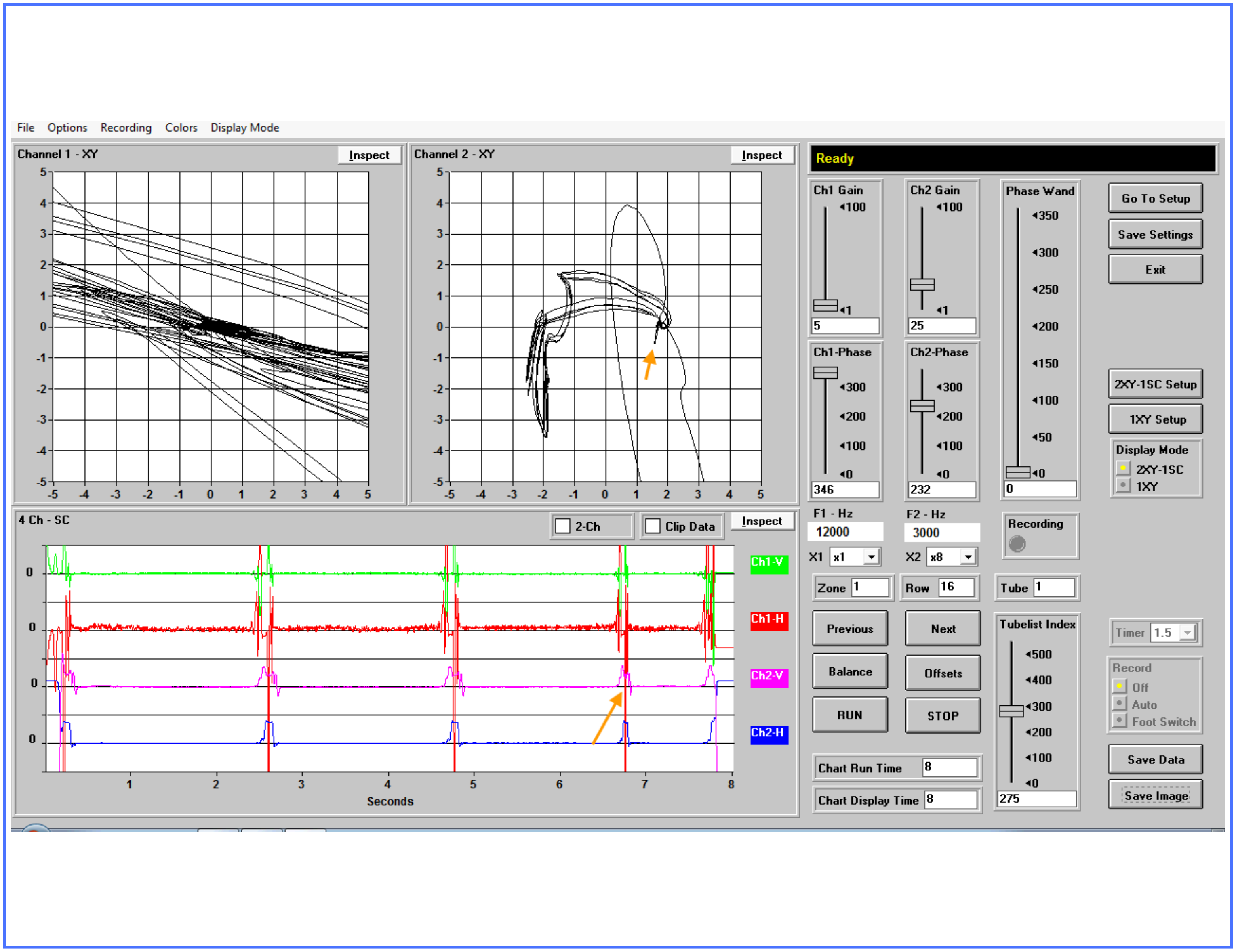
S/N YMXM604029

Condenser Section



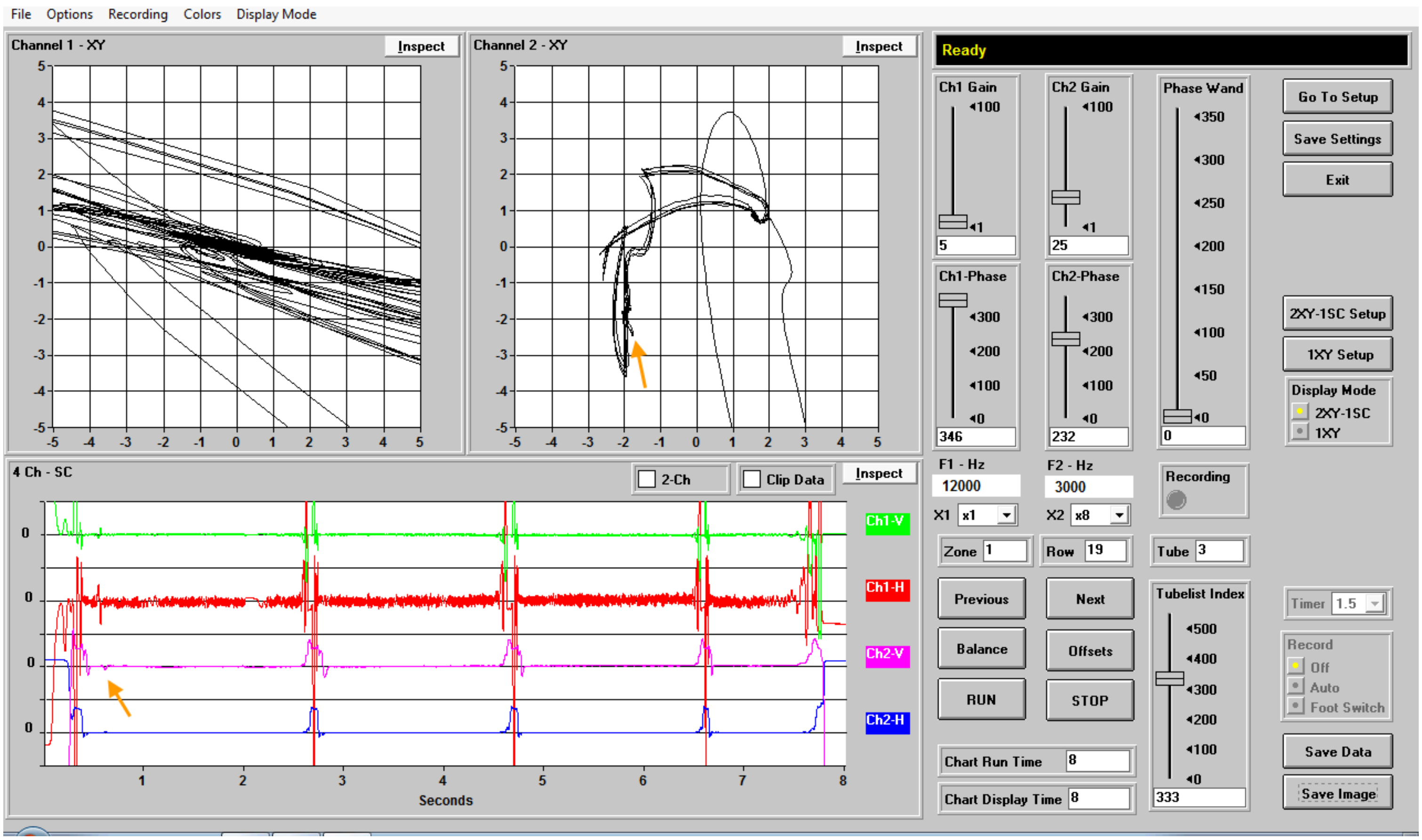
TYPICAL GOOD TUBE (Row 1 Tube 1)

Condenser Section



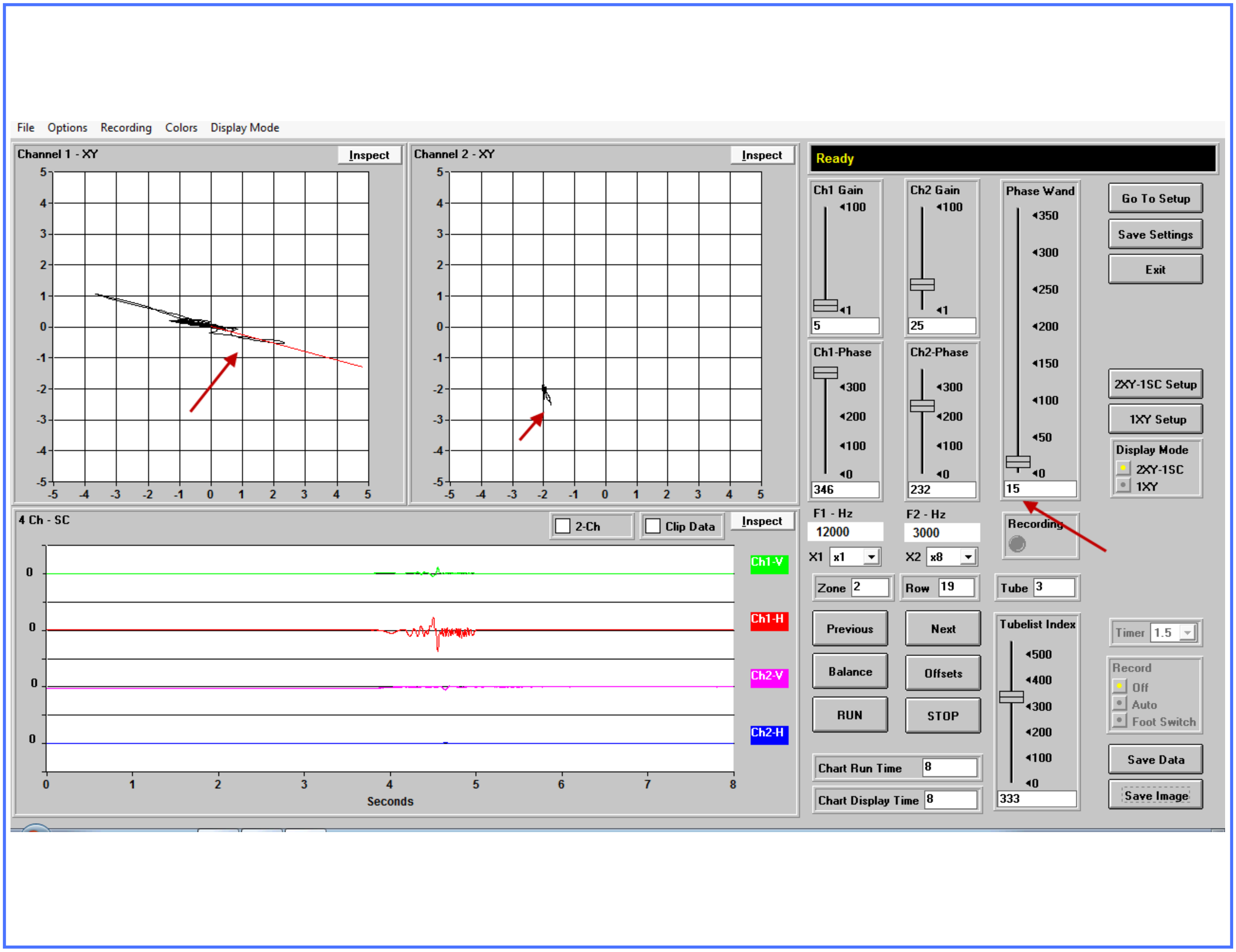
ODML@S < 10%   (Row 16 Tube 1)

Condenser Section



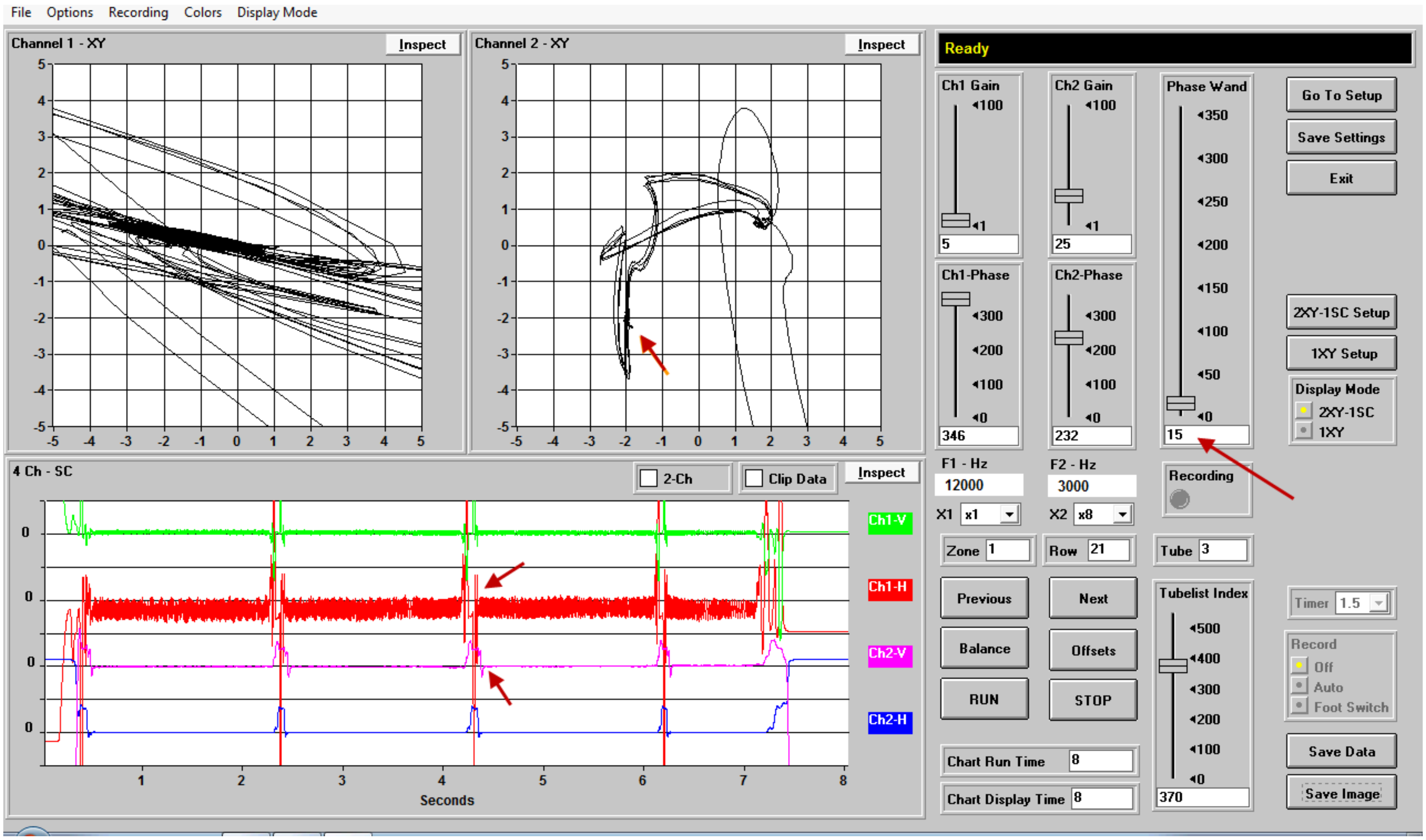
PLF 40% - 59% (Row 19 Tube 3)

Condenser Section



PLF 40% - 59% (Row 19 Tube 3)

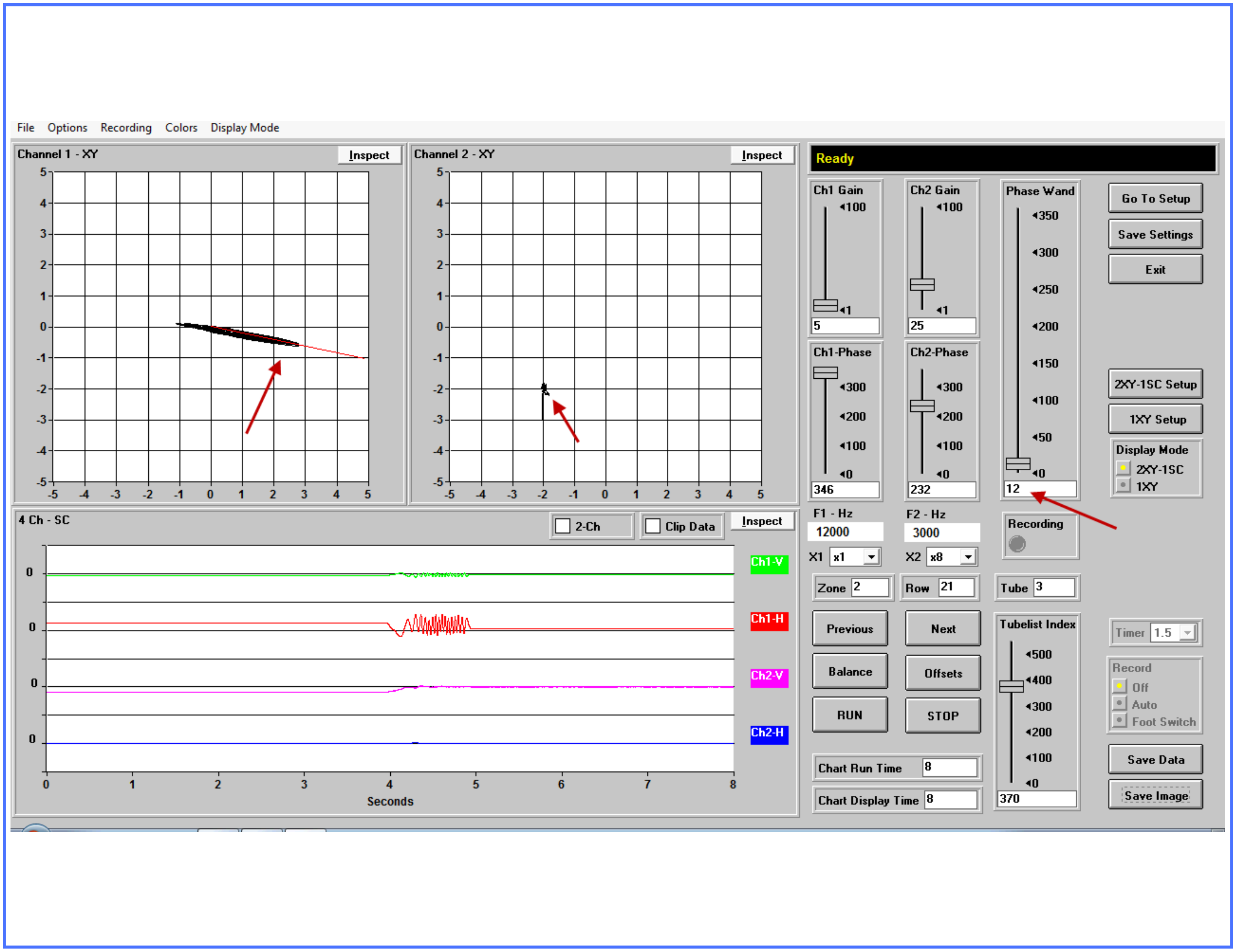
Condenser Section



PLF 40% - 59%   (Row 21 Tube 3)

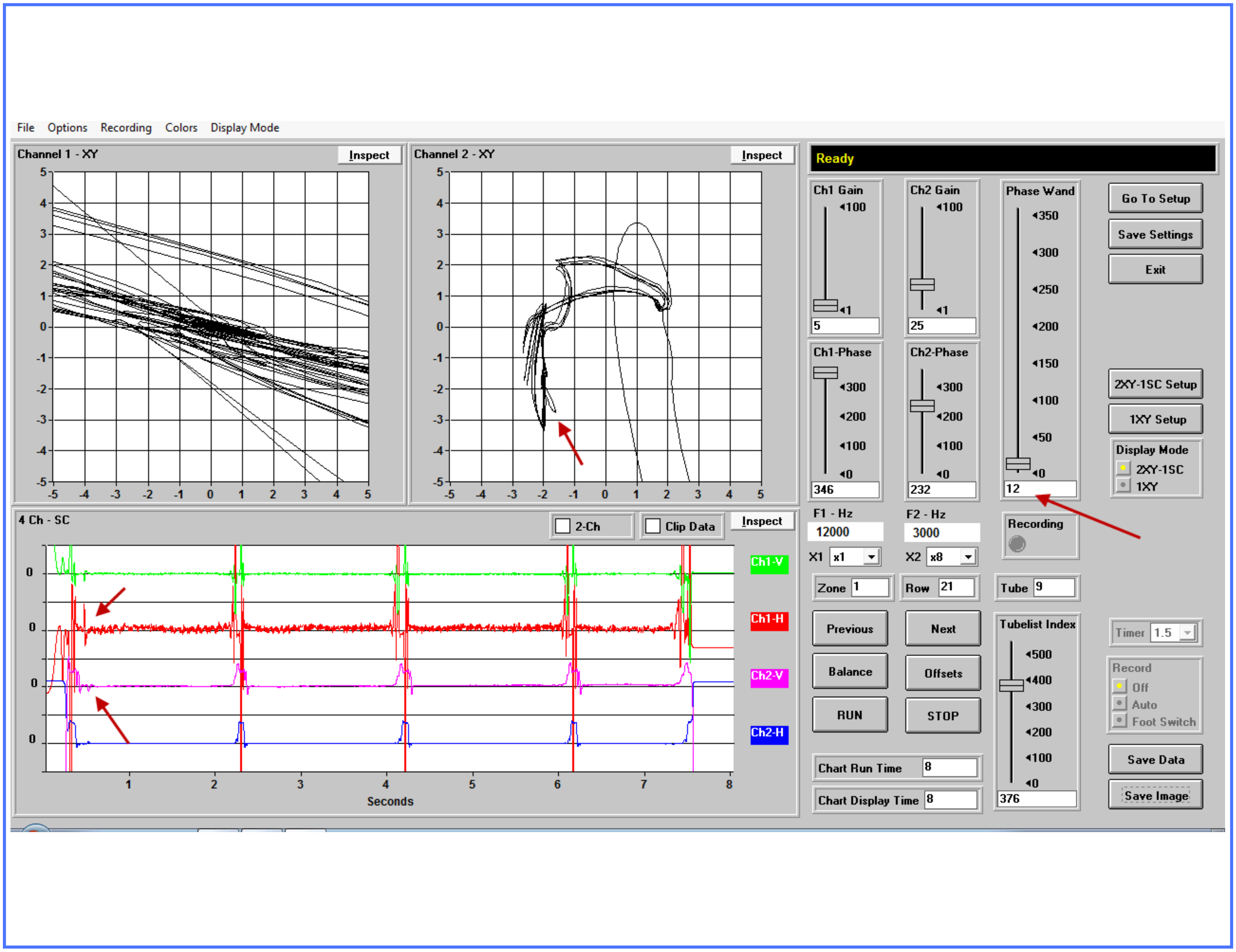
S/N YMXM604029

Condenser Section



PLF 40% - 59% (Row 21 Tube 3)

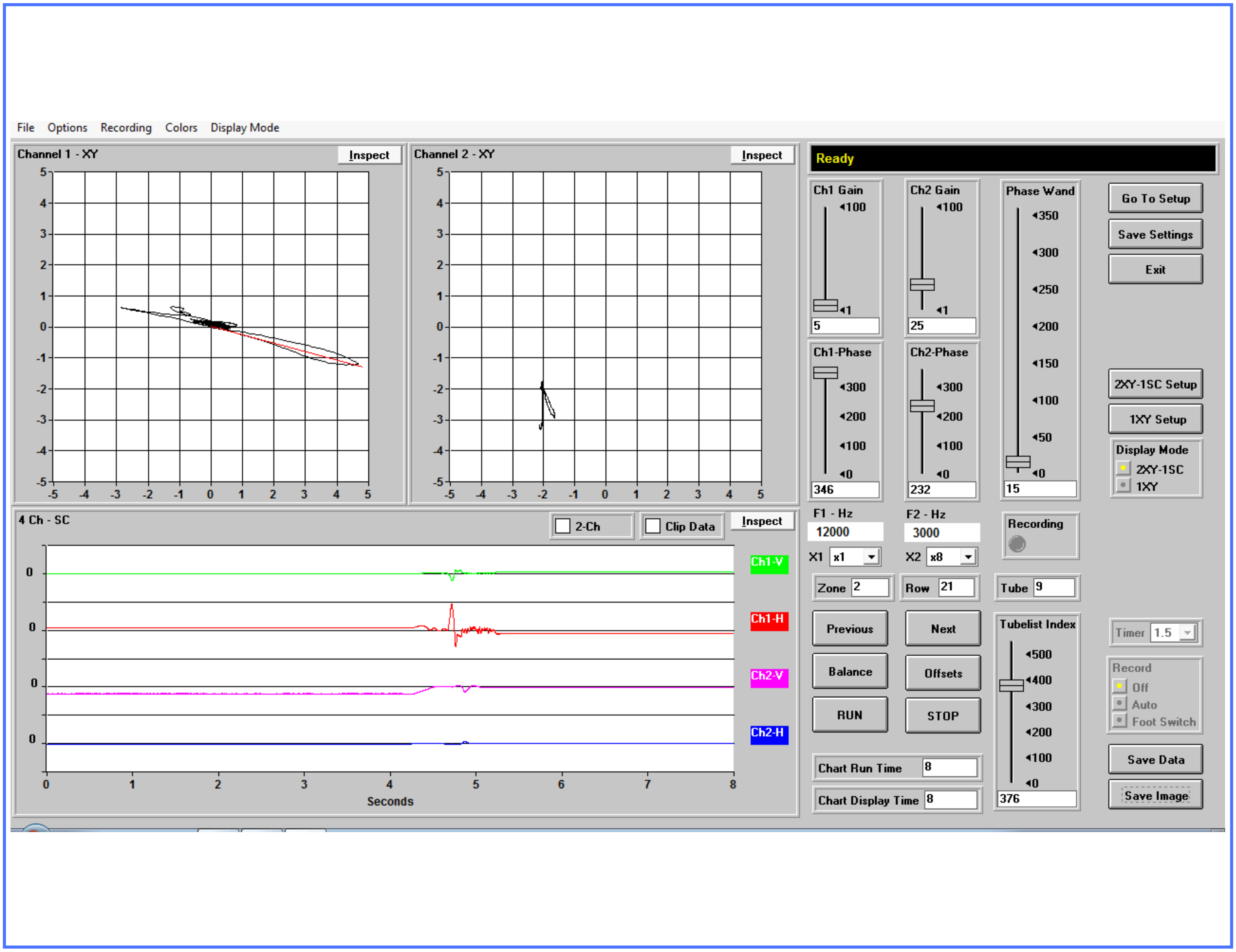
Condenser Section



PLF 40% - 59% (Row 21 Tube 9)

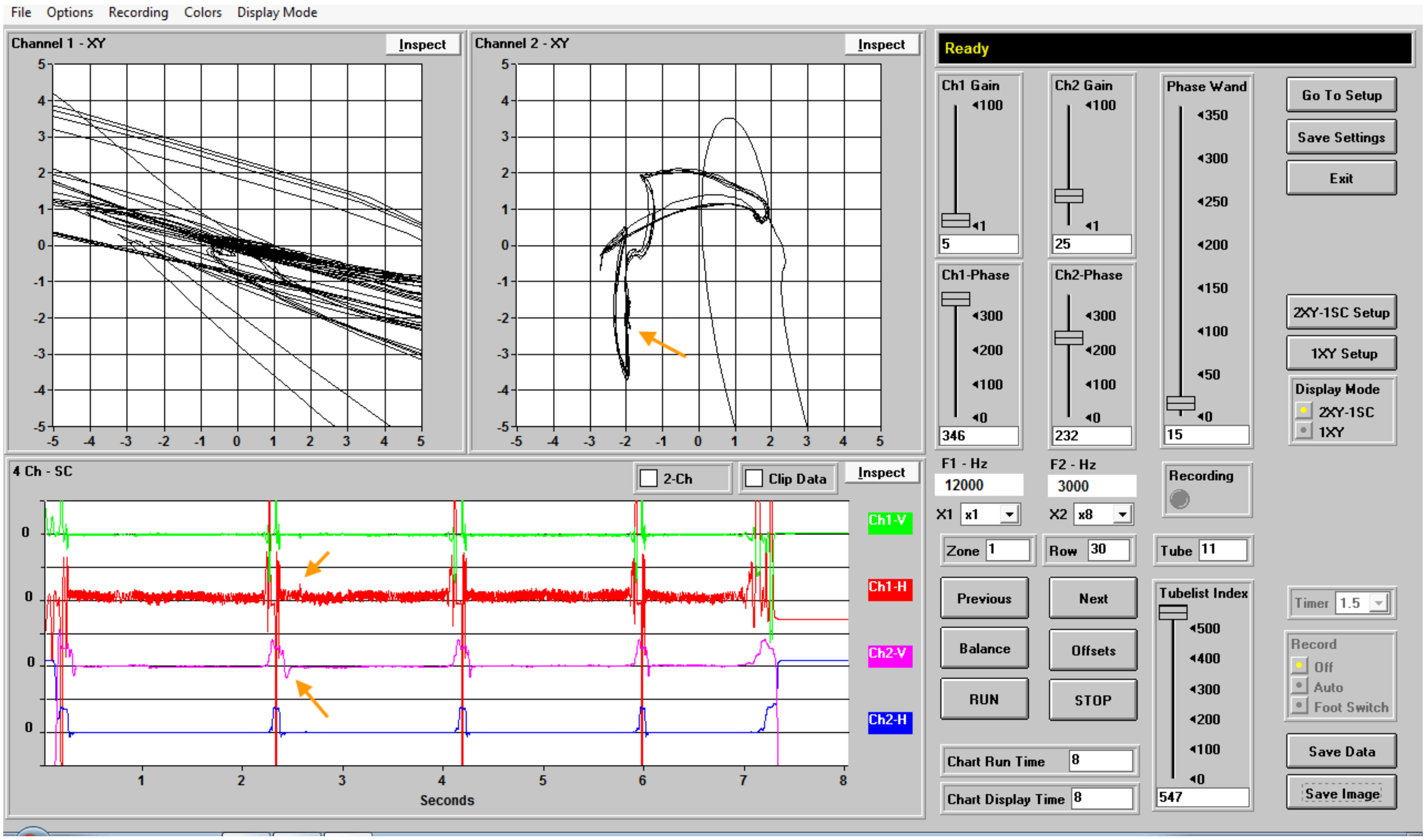
S/N YMXM604029

Condenser Section



PLF 40% - 59% (Row 21 Tube 9)

Condenser Section



PLF 20% - 39% (Row 30 Tube 11)

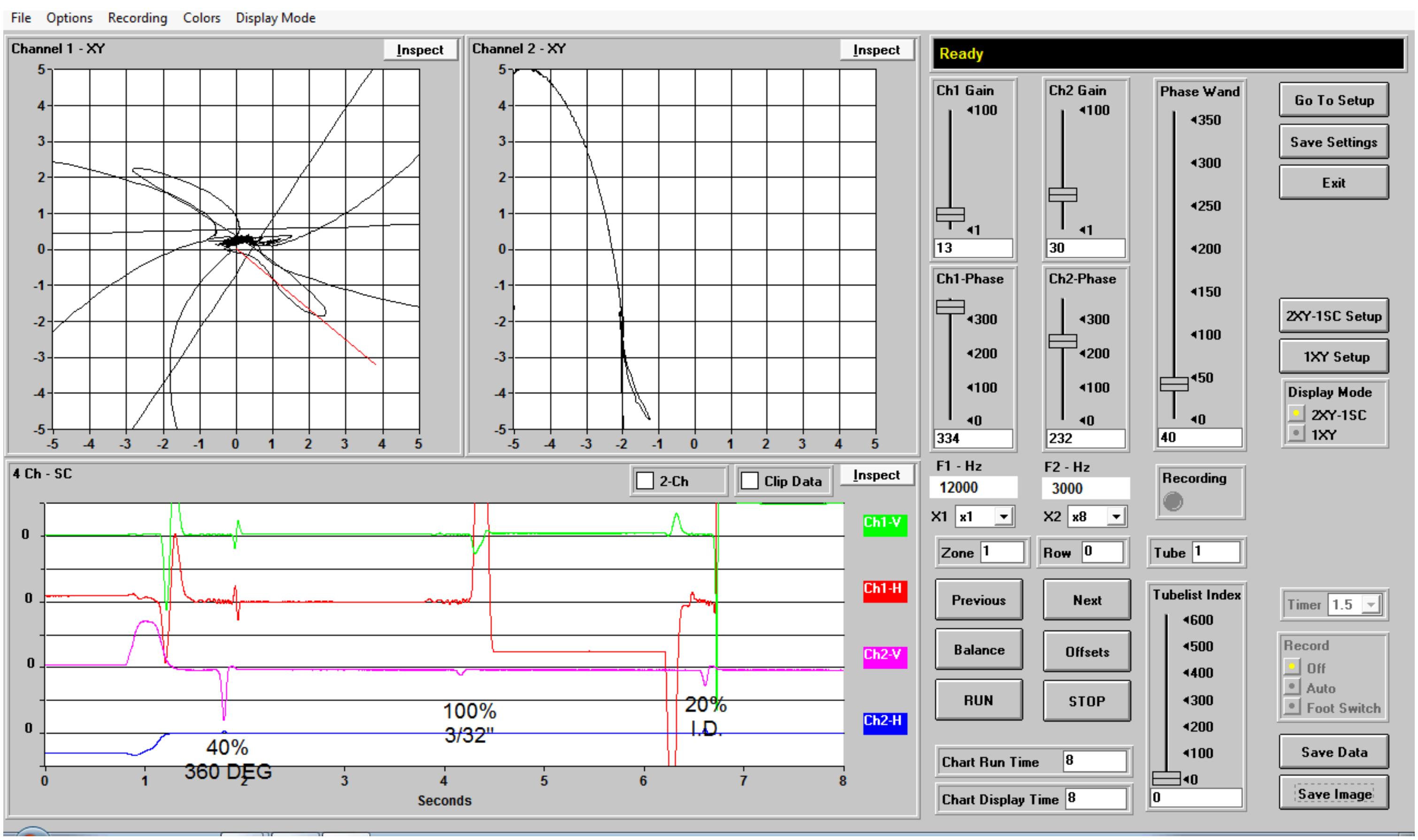
Test Instrument Used
Highline Automation, EC14

S/N YMXM604029

Calibration Page

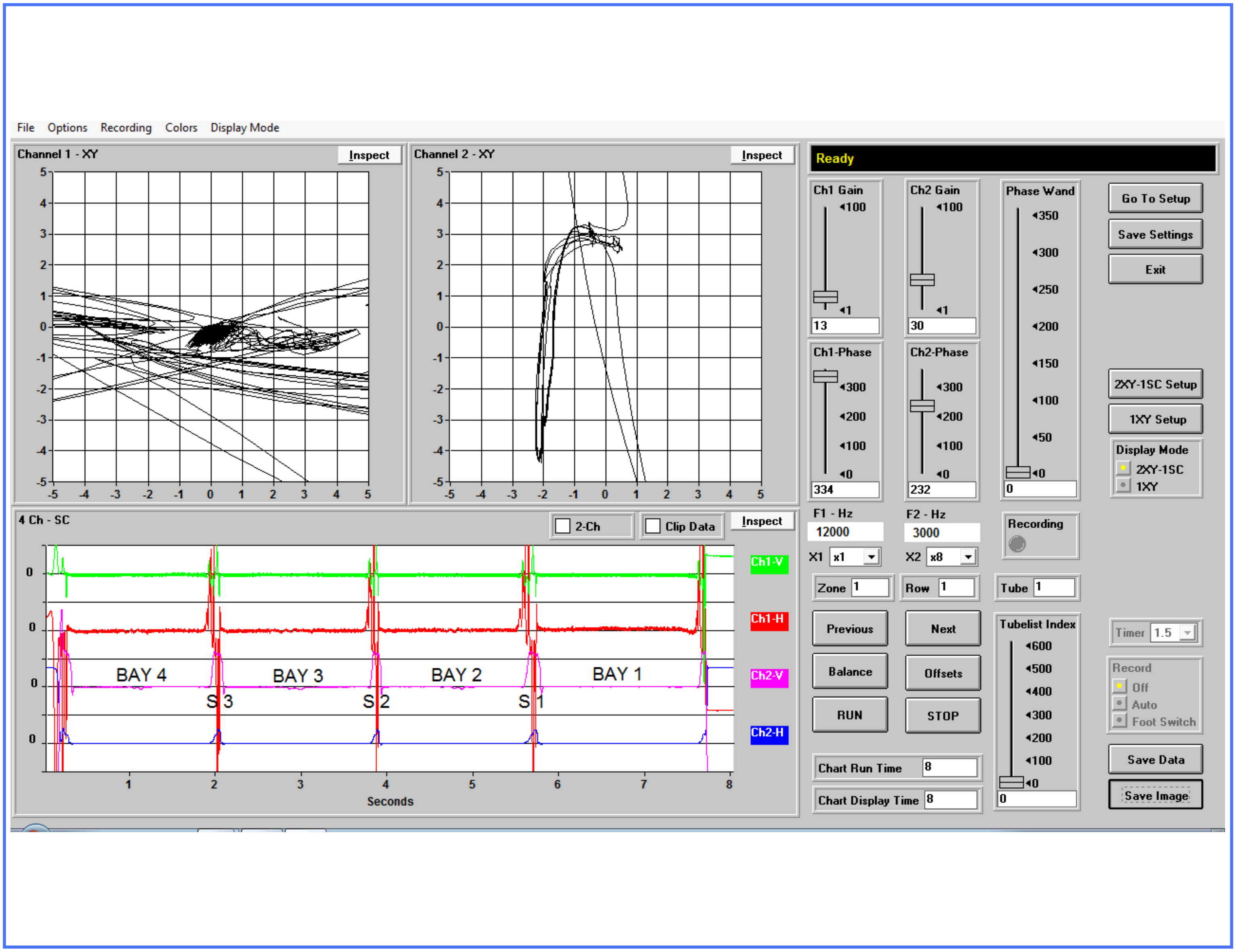
Tube Type	Material	Nom Wall Thick	End Wall Thick	OD	Test Type	Probe Diameter
Skip Fin IE	Copper	.028	.049	.750	CROSS/DIFF	.5625

Evaporator



Note: Defects are compared to machined standards.
Actual Defect Geometry may differ.

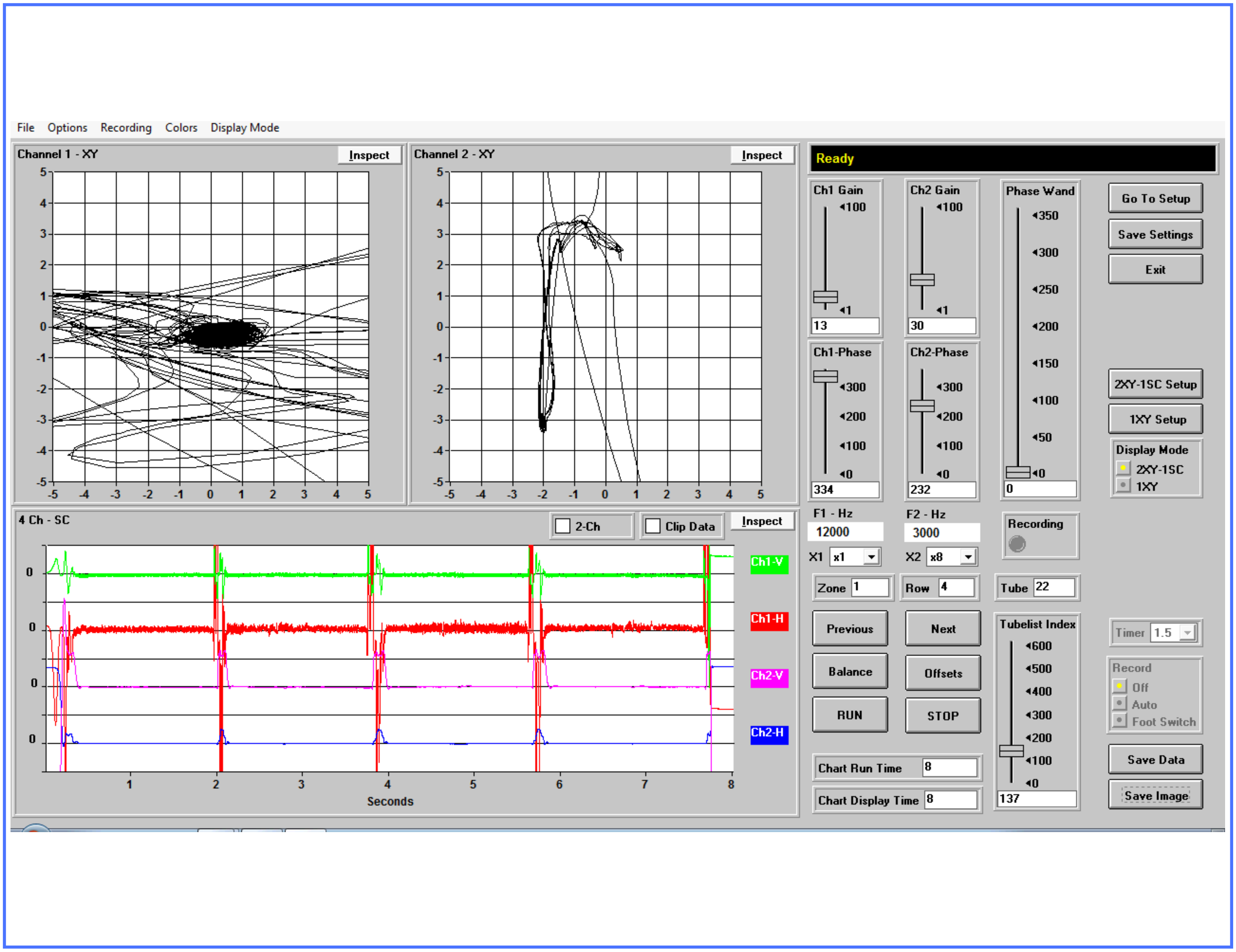
Evaporator Section



TYPICAL GOOD TUBE (Row 1 Tube 1)

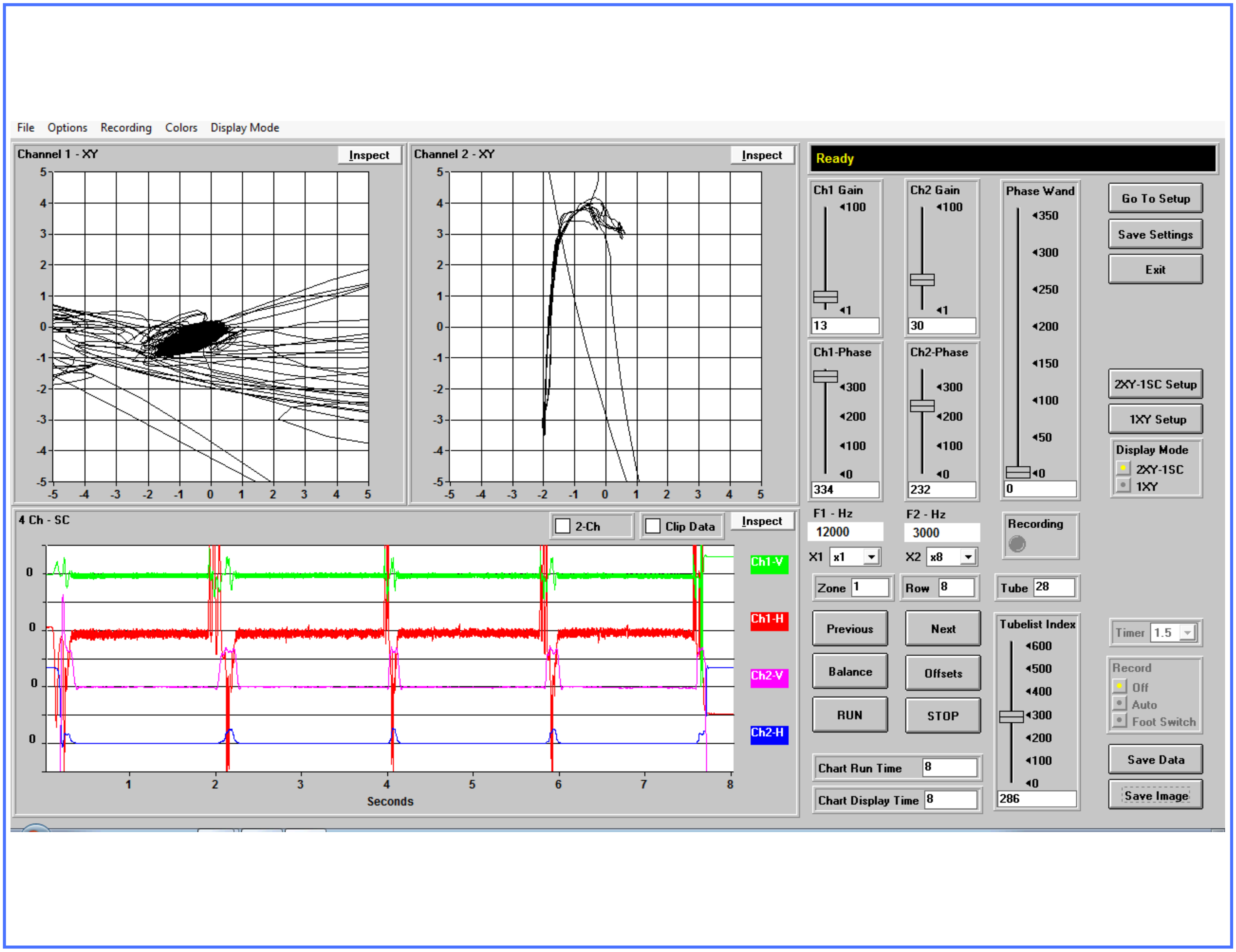
S/N YMXM604029

Evaporator Section



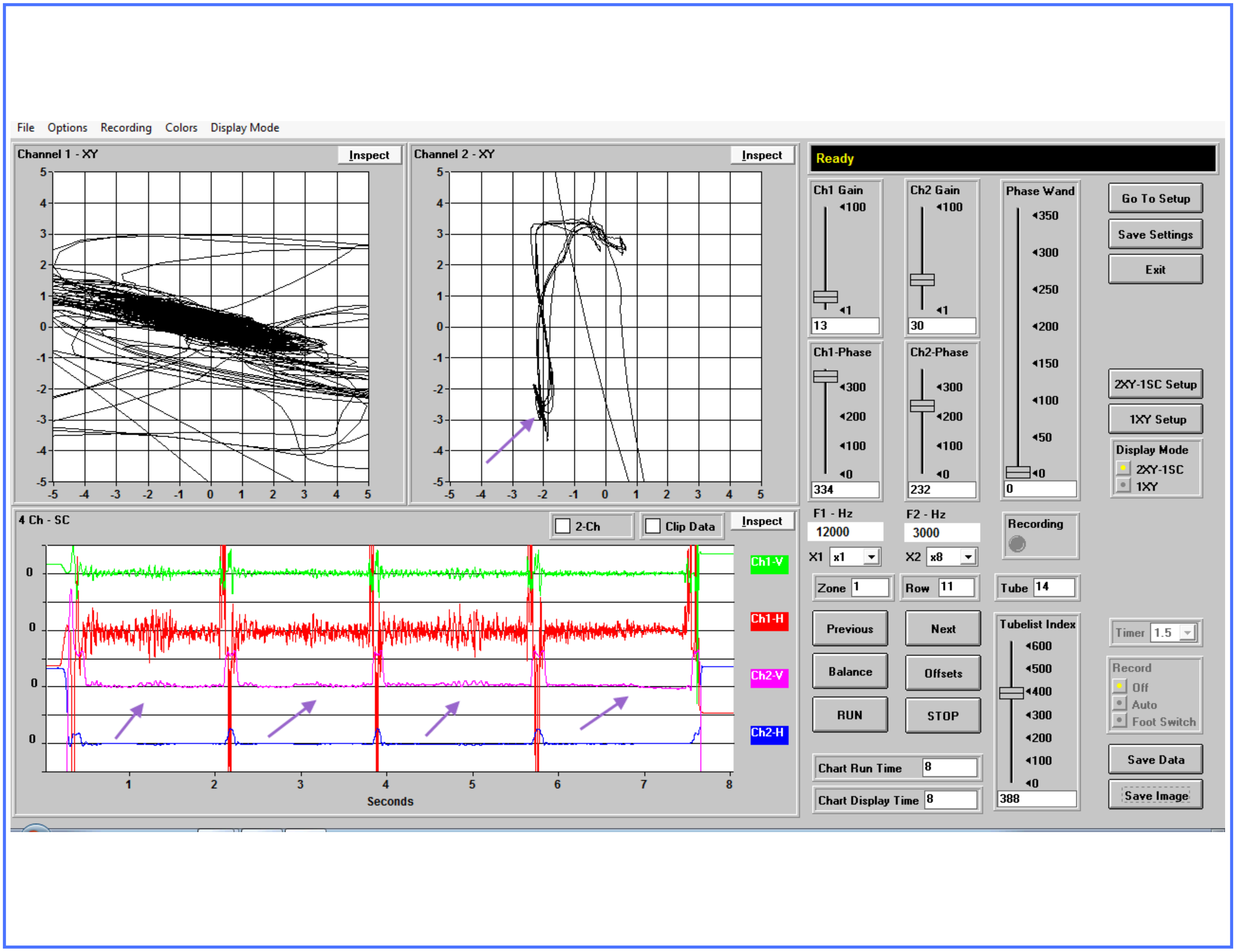
NO SIGNIFICANT DEFECTS (Row 4 Tube 22)

Evaporator Section



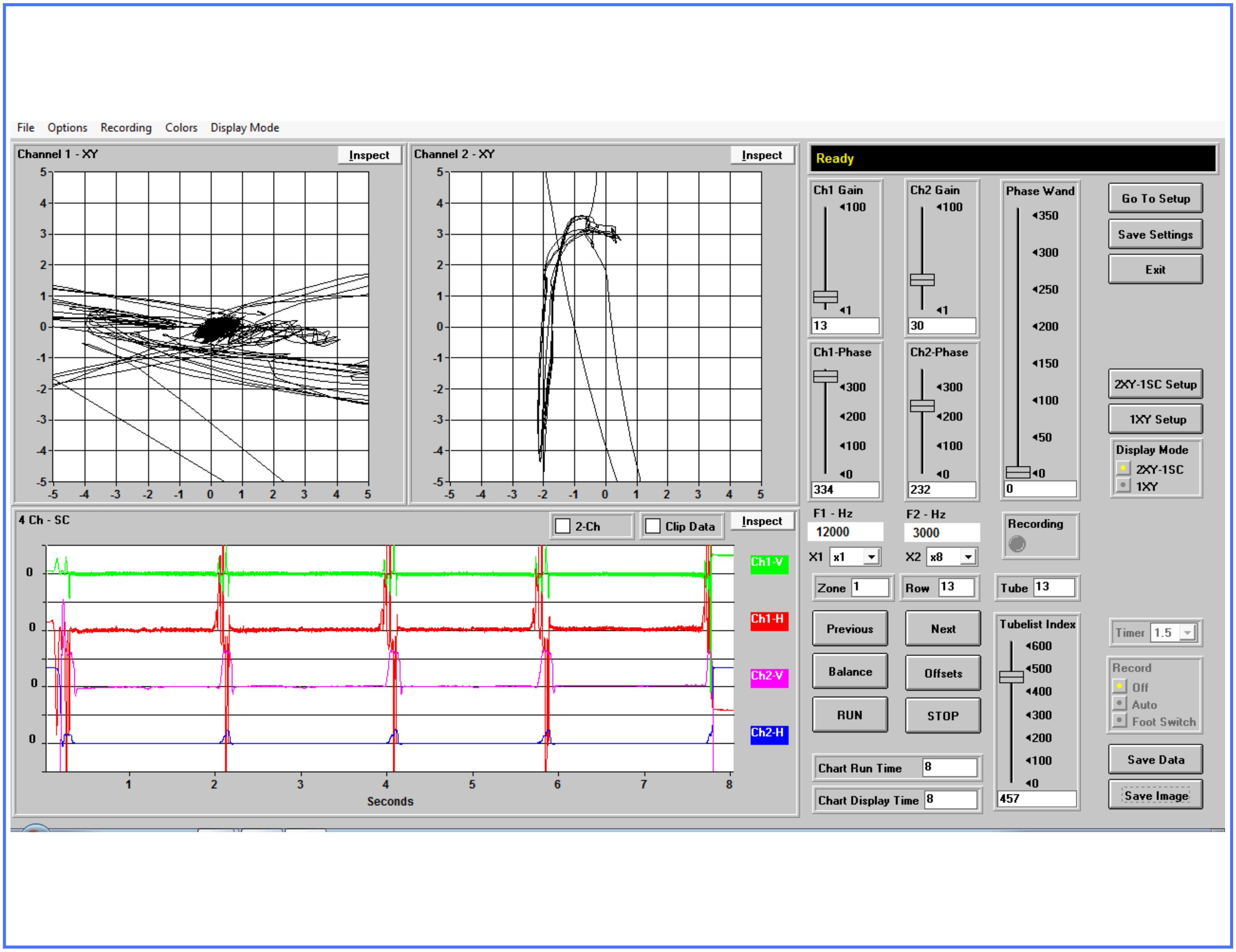
NO SIGNIFICANT DEFECTS (Row 8 Tube 28)

Evaporator Section



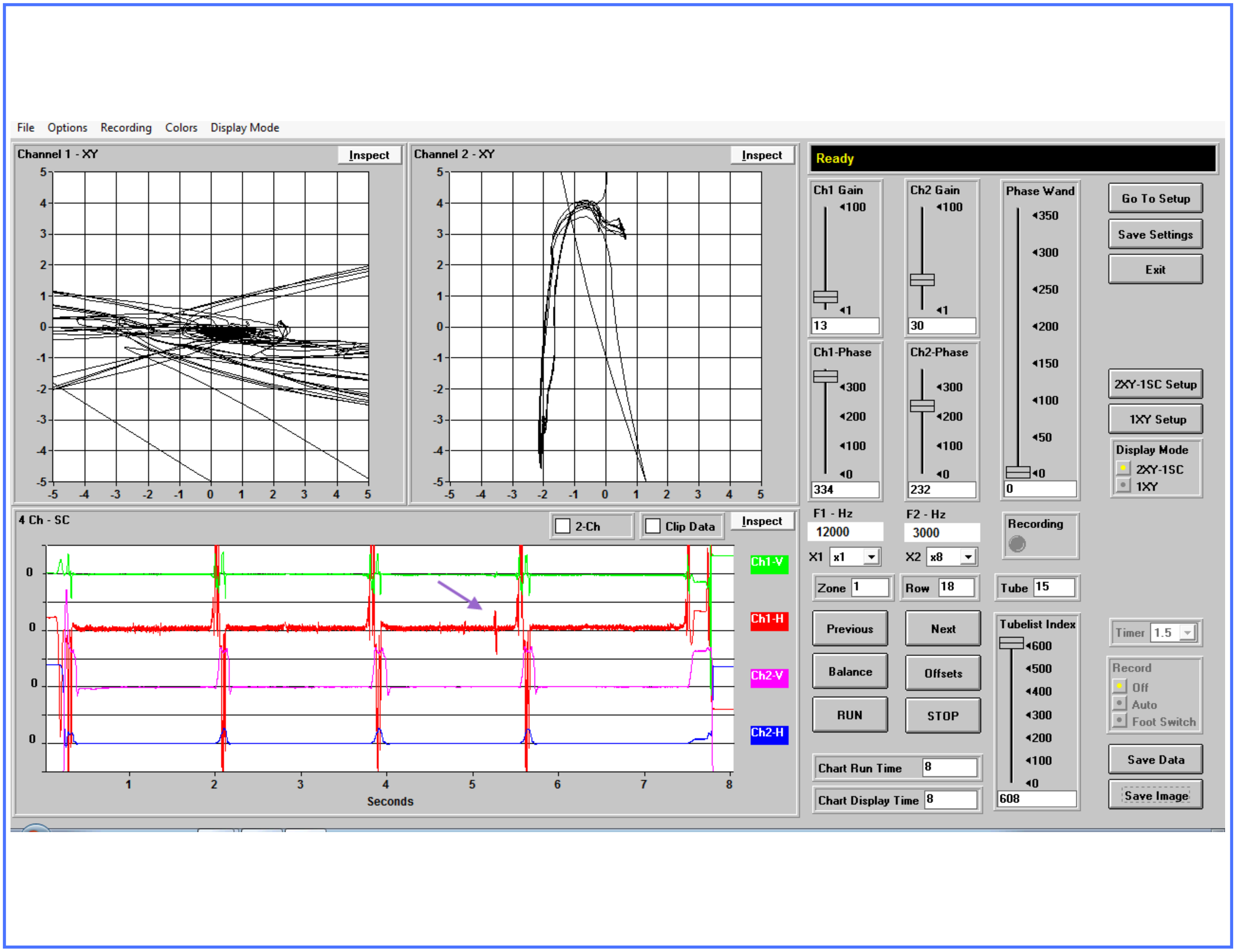
MFG ANOMALY (Row 11 Tube 14)

Evaporator Section



NO SIGNIFICANT DEFECTS (Row 13 Tube 13)

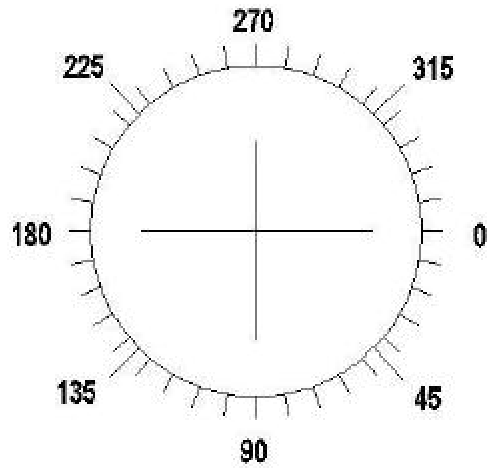
Evaporator Section

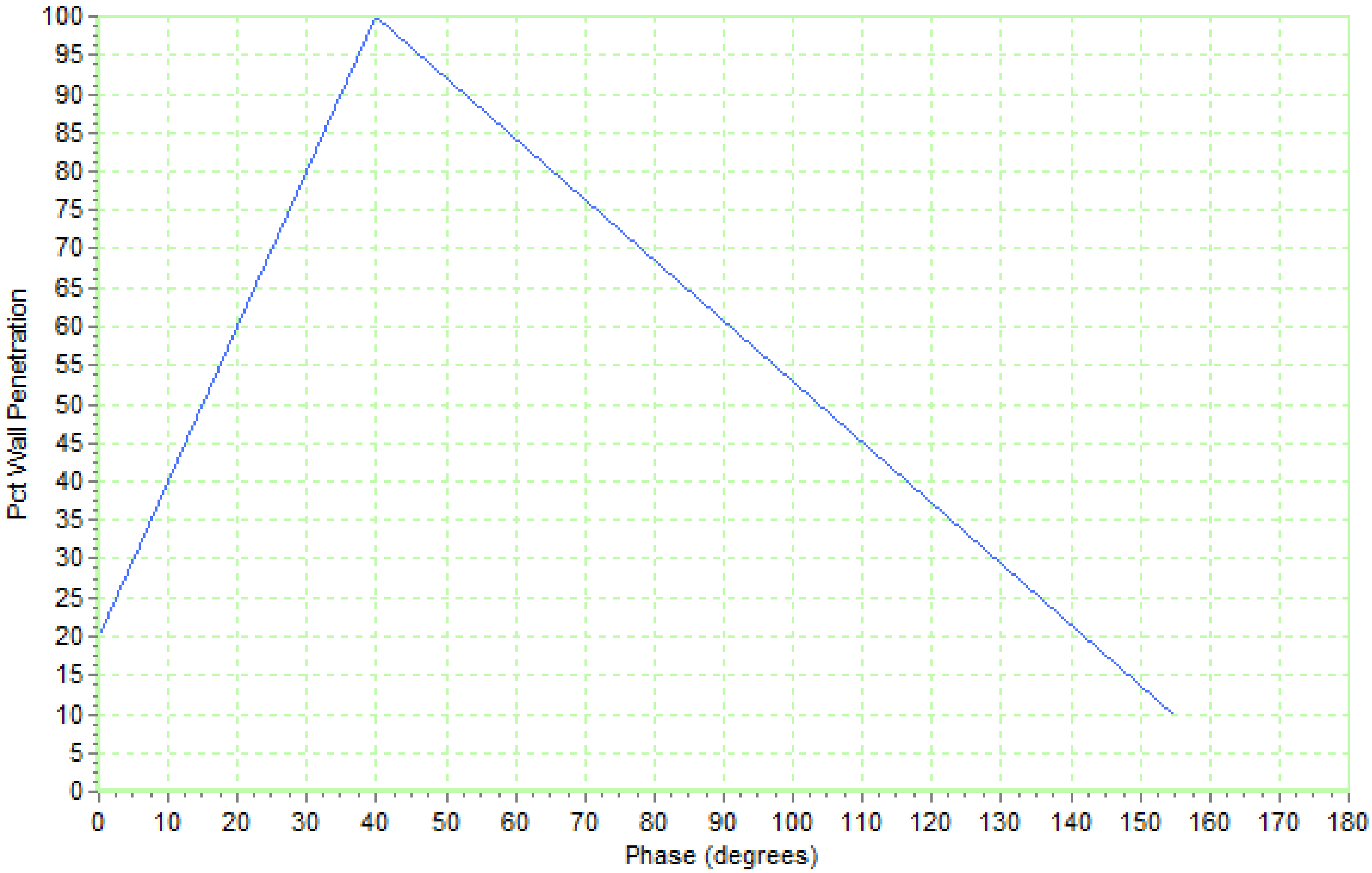


MFG ANOMALY (Row 18 Tube 15)

Phase Chart - Condenser

Material	Tube Type	OD	Wall	Test Type	Frequency	Probe Diameter
Copper	Skip Fin IE	.750	.035	CROSS/DIFF	12000	.500

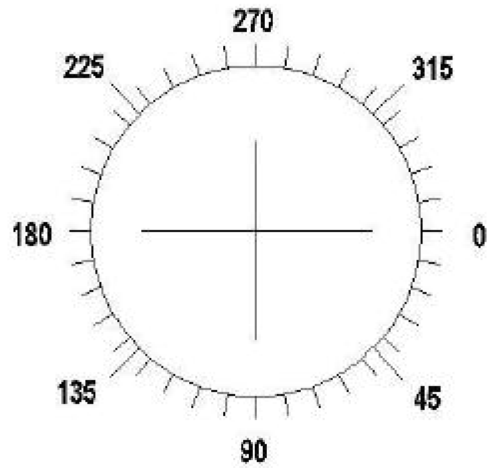


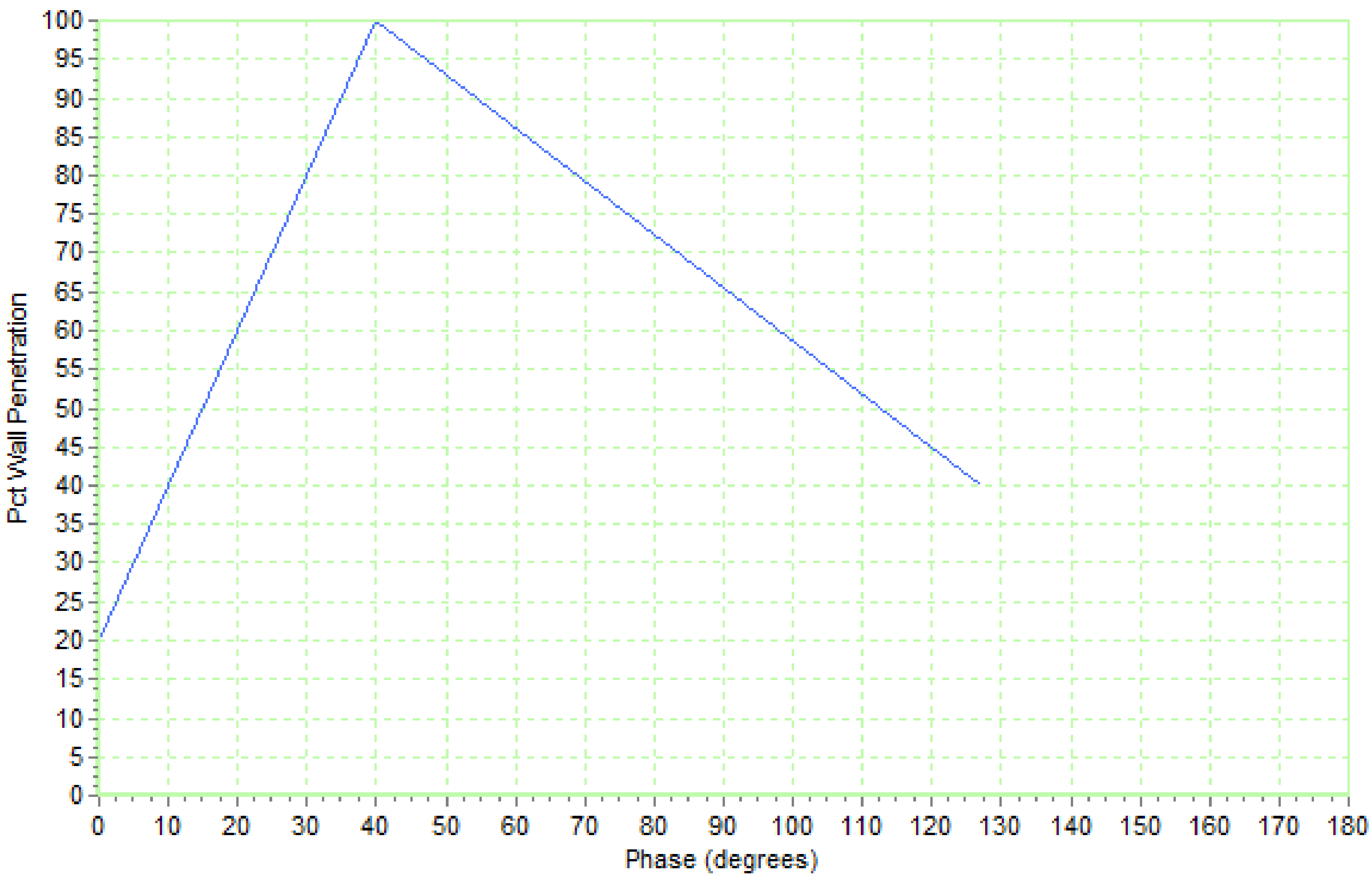


S/N YMXM604029

Phase Chart - Evaporator

Material	Tube Type	OD	Wall	Test Type	Frequency	Probe Diameter
Copper	Skip Fin IE	.750	.028	CROSS/DIFF	12000	.5625





Explanation of Abbreviations

Abbreviation	Explanation
ABN IND	Abnormal Indication
B	Bay
FB	Freeze Bulge
FBH	Flat Bottom Hole
FM	Foreign Material
ID	Internal Diameter
ID CORROSION	Internal Diameter, Corrosion
ID DEPOSIT	Internal Diameter, Deposit
ID PIT	Internal Diameter, Pit
IDML	Internal Diameter, Metal Loss
IE	Internally Enhanced
OD	Outside Diameter
ODML	Outside Diameter, Metal Loss
ODML@S	Outside Diameter Metal Loss at Support
OD DEPOSIT	Outside Diameter, Deposit
PLF	Possible Longitudinal Flaw
PRF	Possible Radial Flaw
PSC	Possible Stress Corrosion
S	Support
WAS	Wear at Support
>	Greater Than
<	Less Than
OTE	Opposite Test End
TE	Test End

Calibration Procedure

A calibration procedure is performed prior to an inspection, and is repeated every 2 hours, or whenever improper operation of the test instrument is suspected. Test frequencies are selected prior to an inspection through experimentation to achieve optimum phase separation, and amplitude response for the tube type and alloy being inspected. An appropriate inspection probe is selected based on tube type, wall thickness, and alloy. The inspection probe will have a minimum fill factor of 80% through the smallest areas of the tubes being inspected. Instrument sensitivity is set high enough to determine background noise inherent in the tube and to produce a .05 Volt deflection for a .031 through wall hole at .25 V/Div.

Calibration Reference Standard

A Calibration Reference Standard representing a typical production run tube of the same alloy, tube type and nominal wall thickness is used to adjust test system response. The calibration reference standard used for the inspection of finned and internally enhanced tubing, has been milled in accordance with the American Society for Testing and Materials (ASTM). Standard Recommended Practices, E-243-80, E-426-76, and E571-76. The depth of the grooves and notches used for establishing instrument response are calculated to compensate for the influence of the fins and/or internal enhancements used on finned tubes. Where applicable, calibration reference standards are milled in accordance with the American Society of Mechanical Engineers (ASME), Section V, Article 8, Appendix I.

A strip chart recording of each calibration reference standard used for the inspection has been included in this report. Each artificial discontinuity has been identified on the strip chart recording.